

The Solar System: An Overview

CSE 4316 — Study Assistant Demo Document

1. Introduction

The Solar System consists of the Sun and all the celestial objects gravitationally bound to it. This includes eight planets, five officially recognized dwarf planets, numerous moons, asteroids, comets, and interplanetary dust. The Solar System formed approximately 4.6 billion years ago from the gravitational collapse of a giant molecular cloud.

2. The Sun

The Sun is a G-type main-sequence star (G2V) located at the center of our Solar System. It contains 99.86% of the total mass of the Solar System. The Sun's core temperature reaches approximately 15 million degrees Celsius, where nuclear fusion converts hydrogen into helium, releasing enormous amounts of energy. The surface temperature of the Sun is about 5,500 degrees Celsius.

Key Facts about the Sun:

- Diameter: 1,392,700 km (109 times Earth's diameter)
- Mass: 1.989×10^{30} kg
- Age: approximately 4.6 billion years
- Distance from Earth: 149.6 million km (1 Astronomical Unit)
- Composition: 74% hydrogen, 24% helium, 2% other elements

3. The Eight Planets

The Solar System has eight recognized planets divided into two groups: the inner rocky planets (terrestrial planets) and the outer gas/ice giants.

Inner Planets (Terrestrial):

Mercury: Smallest planet, closest to the Sun. No atmosphere, extreme temperature swings from -180 to 430 degrees Celsius. Orbital period: 88 Earth days.

Venus: Hottest planet due to runaway greenhouse effect. Surface temperature: 465 degrees Celsius. Rotates in the opposite direction to most planets. Orbital period: 225 Earth days.

Earth: Only known planet with life. Has liquid water, a protective magnetic field, and a nitrogen-oxygen atmosphere. One natural satellite: the Moon. Orbital period: 365.25 days.

Mars: Called the Red Planet due to iron oxide on its surface. Home to Olympus Mons, the largest volcano in the Solar System. Two small moons: Phobos and Deimos. Orbital period: 687 Earth days.

Outer Planets (Gas and Ice Giants):

Jupiter: Largest planet in the Solar System. Has 95 known moons including Ganymede (largest moon in Solar System). Famous for the Great Red Spot storm. Orbital period: 11.9 Earth years.

Saturn: Known for its spectacular ring system made of ice and rock. Second largest planet. Has 146 known moons. Density is so low it could float on water. Orbital period: 29.5 Earth years.

Uranus: Ice giant that rotates on its side (axial tilt of 98 degrees). Has 13 known rings and 28 moons. Coldest planetary atmosphere at -224 degrees Celsius. Orbital period: 84 Earth years.

Neptune: Farthest planet from the Sun. Has the strongest winds in the Solar System reaching 2,100 km/h. Has 16 moons including Triton which orbits in reverse. Orbital period: 165 Earth years.

4. Dwarf Planets

The International Astronomical Union (IAU) officially recognizes five dwarf planets: Pluto, Eris, Haumea, Makemake, and Ceres. Pluto was reclassified from a planet to a dwarf planet in 2006. Ceres is the only dwarf planet located in the asteroid belt between Mars and Jupiter.

5. The Asteroid Belt

The asteroid belt is a region of the Solar System located between the orbits of Mars and Jupiter. It contains millions of irregularly shaped rocky bodies called asteroids. The total mass of all asteroids in the belt is less than the mass of the Moon. The largest object in the asteroid belt is Ceres, which is classified as a dwarf planet.

6. Comets and the Kuiper Belt

Comets are small icy bodies that develop a visible tail when close to the Sun. They originate from two regions: the Kuiper Belt (30-50 AU from the Sun) and the Oort Cloud (2,000-100,000 AU). The most famous comet is Halley's Comet, which returns to the inner Solar System approximately every 76 years. Its last perihelion was in 1986 and the next is expected in 2061.